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Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

[Conventions, stated once here and used throughout: Gaussian units for electrodynamics, SI for mechanics, $c = 1$. Metric signature and the definition of u^α, a^α to be fixed here. Note that the 2006 text declared Gaussian units but evaluated τ_0 without the factor $1/4\pi\varepsilon_0$. The corrected value is used here from the start.]

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge, Maxwell's equations, and the Lorentz force law (Z. Zhang 1979). Maxwell's equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell's equations say how that self-field is generated but not how it acts back. The Lorentz force law says how a field acts on a charge but is silent on whether the charge's own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it. But the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart. Bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. H. Poincaré (1906) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–energy tensor a binding tensor called the Poincaré stresses, chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence. In the century since, few have followed it.

H. A. Lorentz (1909) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of Sec. 1.3.

P. A. M. Dirac (1938) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow. It is set out in Sec. 2.

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform. The model carries no internal structure that has to be specified in advance. The equation of motion comes out in a fixed form. The construction also generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so the force has to be extracted indirectly. This can be done either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz's electron is a sphere of finite radius carrying a uniform charge. It is far enough into the extended picture that nothing diverges, and structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton's third law. Nothing is genuinely lost. The missing momentum resides in the field. But the failure is a fair measure of how far the problem has already moved away from mechanics.

1.3.1. *Deriving the self-force directly*

The calculation itself is not deep. The work is all in the expansion, and it is standard (see, e.g., J. D. Jackson 1999; F. Rohrlich 2007). Let the charge be distributed with density ρ over a rigid body of size R, moving slowly enough that magnetic contributions may be dropped to the order kept. The self-force is the double integral of the retarded interaction over all pairs of charge elements,

$$\mathbf{F}_{\text{self}} = \iint d^3r\, d^3r'\, \rho(\mathbf{r})\rho(\mathbf{r}')\, \mathbf{K}(\mathbf{r}, \mathbf{r}';\, t - s)\,, \qquad s = \frac{|\mathbf{r} - \mathbf{r}'|}{c}, \tag{1}$$

in which the kernel $\mathbf{K}$ is evaluated not at the present time but one light-crossing time s earlier. Since $s \sim R/c$ is small, expand every retarded quantity about the present instant,

$$f(t - s) = f(t) - s\,\dot{f}(t) + \frac{s^2}{2}\ddot{f}(t) - \frac{s^3}{6}\dddot{f}(t) + \cdots\,, \tag{2}$$

and collect. Each additional power of s costs one factor of R/c and buys one more time derivative of $\mathbf{v}$. The series therefore organizes itself by derivative order. The term containing $d^n\mathbf{v}/dt^n$ carries R^{n-2}. The zeroth term is the static self-repulsion, which vanishes by symmetry for a rigid symmetric body, and what survives is

$$\mathbf{F}_{\text{self}} = -\frac{4U_0}{3c^2}\,\dot{\mathbf{v}} \;+\; \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}} \;+\; \sum_{n\geq 3} \alpha_n\,\frac{e^2 R^{n-2}}{c^{n+1}}\frac{d^n\mathbf{v}}{dt^n}, \tag{3}$$

where U_0 is the electrostatic self-energy and the α_n are pure numbers fixed by the shape of the distribution.

The three groups of terms meet three quite different fates, set out in Table 1, and the whole of the subject descends from that split.

Term	Coefficient scales as	What becomes of it
$n = 1$, $\propto \dot{\mathbf{v}}$	R^{-1}, divergent	absorbed into the mass
$n = 2$, $\propto \ddot{\mathbf{v}}$	R^0, model-independent	survives as the radiation reaction
$n \geq 3$, $\propto d^n\mathbf{v}/dt^n$	R^{n-2}, vanishing	lost in the point limit

Table 1. The three groups of terms in the expansion Eq. (3), and what becomes of each as the radius of the body is taken to zero. Only the middle row survives, and the classical theory of the electron is the theory of that one row.

The first row is the divergence, and it is harmless only because of its form. Being proportional to $\dot{\mathbf{v}}$, it cannot be distinguished from mass times acceleration. It can therefore be hidden in the mass,

$$m = m_{\text{bare}} + \frac{4U_0}{3c^2}. \tag{4}$$

This is the first appearance in the subject of what would later be called mass renormalization, and the source of the first defect below.

The second row is the one that matters, and its coefficient cannot depend on how the charge is arranged. With no power of R available there is nothing left to build a coefficient from except e and c, so the combination is fixed up to a pure number. A sphere, a shell and a dumbbell all return $2e^2/3c^3$, while each returns its own coefficient for the divergent term. That asymmetry is the reason the radiation-reaction term is believed and the mass term is renormalized away.

The third row is the casualty. Those terms are the only place where the internal structure of the electron would ever have shown itself, and the point limit destroys precisely that information.

Substituting Eq. (3) into $m_{\text{bare}}\dot{\mathbf{v}} = \mathbf{F} + \mathbf{F}_{\text{self}}$ and using Eq. (4) gives

$$\boxed{m\dot{\mathbf{v}} - \frac{2e^2}{3c^3}\ddot{\mathbf{v}} + \cdots = \mathbf{F}} \tag{AL}$$

now usually called the Abraham–Lorentz equation (M. Abraham 1905; H. A. Lorentz 1909), in which m is the observed mass and everything from the second term on the left is the self-force.

1.3.2. *The same result from energy balance*

There is a much shorter route, which avoids the charge distribution altogether and therefore never meets the divergent mass. Take the Larmor formula (J. Larmor 1897) for the radiated power, $P = (2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}$, and demand that the work done by the self-force account for the energy radiated over an interval,

$$\int_{t_1}^{t_2} \mathbf{F}_{\rm self}\cdot\mathbf{v}\,dt = -\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt. \tag{5}$$

Integrating the right-hand side by parts,

$$-\int_{t_1}^{t_2} \frac{2e^2}{3c^3}\dot{\mathbf{v}}\cdot\dot{\mathbf{v}}\,dt = -\frac{2e^2}{3c^3}\Big[\dot{\mathbf{v}}\cdot\mathbf{v}\Big]_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}\cdot\mathbf{v}\,dt. \tag{6}$$

If the boundary term vanishes, as it does for periodic motion or whenever $\dot{\mathbf{v}}\cdot\mathbf{v}$ takes the same value at both ends, the two integrands may be compared:

$$\mathbf{F}_{\rm self} = \frac{2e^2}{3c^3}\,\ddot{\mathbf{v}}, \tag{7}$$

which is the self-force of Eq. (AL) in three lines. The mass never enters, so it stays where classical mechanics put it.

Two things should be said about what this shortcut costs. First, the conclusion is drawn from equality of integrals over a restricted class of intervals, not from any pointwise identity, so it establishes an average effect and not an instantaneous law. The third and higher derivatives of $\mathbf{v}$, which the direct calculation delivers in Eq. (3), are simply invisible to it. Second, and more interestingly, the term thrown away is not junk. The boundary term $(2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\mathbf{v}$ is the Schott energy (G. A. Schott 1912), and the piece of the covariant reaction force whose time derivative it is will turn out to be the Schott term. Everything awkward about energy balance in Sec. 3 is already present here. The radiated power and the work done by the reaction force refuse to match instant by instant. Uniformly accelerated motion radiates while feeling no reaction at all. Both puzzles sit in the term this derivation is allowed to discard because the motion was assumed periodic.

1.3.3. *Three defects*

Three objections can be raised against Eq. (AL).

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The factor 4/3 is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2, and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not. It is the same omission as before, now showing up as a number. E. Fermi (1922) and W. Wilson (1936) later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge. At most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. The energy-balance derivation of Sec. 1.3.2 sidesteps the whole question, which is its chief attraction.

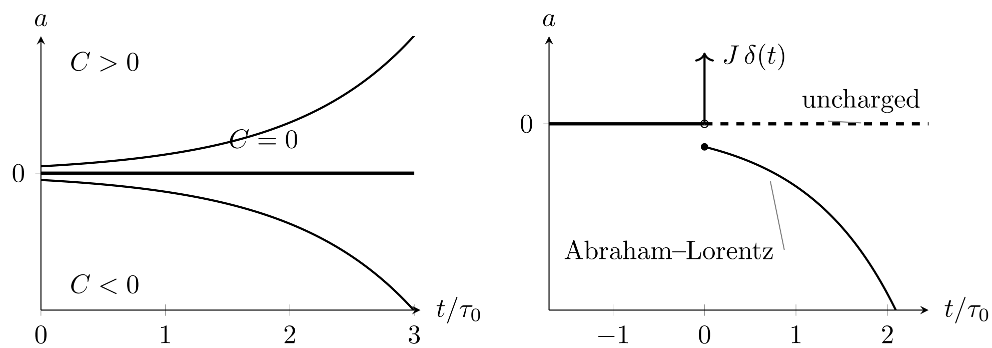
The equation is not covariant. Equation (AL) is written with d/dt in a particular frame. The repair is due to L. D. Landau & E. M. Lifshitz (1975) and is disarmingly short. Replace d/dt by $d/d\tau$ and guess $m\,\dot{u}^\alpha = F^\alpha + k\,\ddot{u}^\alpha$. The guess fails at once, because contracting with u_α and using $u^\alpha u_\alpha = -1$, whose first two derivatives give $u \cdot a = 0$ and $u \cdot \dot{a} = -a \cdot a$, leaves $-k\,(a \cdot a)$ on the right where zero is required. The cure is to add a term along u^α and let the constraint fix its coefficient: demanding $u_\alpha(\dot{a}^\alpha + C\,u^\alpha a \cdot a) = 0$ gives $C = -1$, so that the reaction force must be

$$\frac{2e^2}{3c^3}\left(\dot{a}^\alpha - u^\alpha a^\beta a_\beta\right) = \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta. \tag{8}$$

This is exactly the term that Dirac's far longer construction produces in Sec. 2. Covariance alone, applied to a nonrelativistic result, very nearly hands over the answer. This is worth keeping in mind when judging how much the tube derivation really establishes.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. *The Abraham–Lorentz equation and its runaway solutions*



(a) free electron, $a = C\,e^{t/\tau_0}$ (b) struck by an impulse at $t = 0$

Figure 1. The two faces of the third-order term. *(a)* A free electron obeying Eq. (AL) has $a = C\,e^{t/\tau_0}$. Every $C \neq 0$ runs away; the physical solution is the single value $C = 0$, and nothing in the initial-value problem selects it, since the equation is first order in a and admits any $a(0)$. *(b)* An impulse delivered at $t = 0$. An uncharged particle takes the blow, jumps in velocity, and returns to zero acceleration: the force is over and so is the acceleration. The electron does the opposite. Its acceleration jumps to $\kappa = -J/m\tau_0$ and grows without bound long after nothing is acting on it, and, since κ carries the sign opposite to J, it runs away against the blow that started it. The velocity, by contrast, is continuous at $t = 0$ here: it is the acceleration that jumps.

Set $\mathbf{F} = 0$ in Eq. (AL), keep only the two terms written, and reduce to one dimension. The coefficient of the self-force term is m times a time,

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24}\ \text{s} \quad \text{(electron)}, \tag{9}$$

which will reappear in every section below, and the equation of motion of a free electron collapses to a first-order equation for the acceleration alone,

$$\dot{a} = \frac{a}{\tau_0}, \qquad \text{whence} \qquad a = C\,e^{t/\tau_0}. \tag{10}$$

Both branches of this solution are unsatisfactory, and for different reasons. If $C \neq 0$, an electron with nothing whatever acting on it accelerates itself, and does so on the timescale τ_0, which is to say it gains a factor of e in acceleration every $\sim 10^{-23}$ seconds. In the nonrelativistic equation the velocity grows without bound. In the covariant version of Sec. 3 it approaches c instead, but the character of the solution is unchanged. No intuition about free particles survives this, and no agent is available to be held responsible for the energy.

If $C = 0$ then $a = 0$, the electron moves uniformly, and Newton's first law is recovered. One may say that the Abraham–Lorentz equation tells us that a free electron moves exactly as an uncharged body does. The physical content is unobjectionable. What is objectionable is the standing of the condition. Equation (10) is first order in a and therefore admits any initial value $a(0)$. The physical solution is picked out by setting $a(0) = 0$, and the only reason offered for doing so is that every other choice misbehaves as $t \to \infty$. A condition imposed at infinity to restrict data at $t = 0$ has no warrant in the initial-value problem it is being applied to. This remains true no matter how reasonable the surviving solution looks.

The impulse problem sharpens the objection into something concrete. Write the electron's equation with an impulsive force of strength J delivered at $t = 0$,

$$ma - m\tau_0\,\dot{a} = J\,\delta(t), \tag{11}$$

and compare with the corresponding classical problem. A free mass point struck by an impulse jumps discontinuously in velocity and has zero acceleration immediately afterwards: the blow is over, and so is the acceleration. Solving Eq. (11) with the electron moving uniformly beforehand gives instead

$$a = \begin{cases} 0, & t < 0, \\ \kappa\, e^{t/\tau_0}, & t > 0, \end{cases} \qquad \kappa = -\frac{J}{m\tau_0}. \tag{12}$$

The electron does not stop accelerating when the blow ends. It accelerates ever harder, without limit, long after nothing is acting on it. Since κ carries the opposite sign to J, it runs away in the direction opposite to the blow that started it. This is the runaway, or acceleration collapse, in its starkest form. No choice of initial data avoids it here, because the requirement of uniform motion before the pulse has already been spent.

The root of the behaviour is visible in the order of the equation. Newton's equation of motion contains the acceleration and nothing beyond it, so when the force stops the acceleration stops with it. Equation (AL) contains derivatives of the acceleration, so the state of the electron at a given instant includes how fast its acceleration is changing, and that quantity survives the force that produced it. What sustains it, physically, is the electron's own electromagnetic field. The defect is therefore not an artefact of the sphere model or of the expansion used to reach Eq. (AL). Any equation of motion that accounts for the self-field by adding a derivative of the acceleration will inherit it. Dirac's equation, derived on entirely different principles in Sec. 2, inherits it exactly.

1.5. *Criteria for a physically acceptable equation of motion*

Everything from here on consists of proposals to replace Eq. (AL) with something better behaved, and proposals cannot be judged without a standard to judge them by. It is worth fixing that standard now, before any candidate is on the table, so that it cannot be quietly adjusted to fit whichever equation is under discussion. Three requirements are imposed throughout these notes.

1. **The motion must follow from the equation in the ordinary way.** The particle should obey a differential equation that can be solved by the usual procedure. One supplies initial or boundary data, integrates, and obtains the state at all later times. The data supplied must themselves be data the physical situation could actually provide. This is a weaker demand than it may appear, and Sec. 1.4 has already shown Eq. (AL) failing it: the equation is solvable, but the solution that survives is picked out by a condition imposed at $t \to \infty$, and no experimenter preparing an electron at $t = 0$ has access to that condition.

2. **A stable state must remain stable under a small disturbance.** The concrete test used throughout is the one already applied in Sec. 1.4: take a particle in uniform motion, strike it with a pulse, and ask what it does afterwards. An acceptable equation returns it to uniform motion. It is also acceptable for the particle to settle into a regular variation confined to a bounded range of speeds. This weaker alternative is included deliberately, because Sec. 5 produces solutions of exactly this kind. What is not acceptable is for the particle to accelerate without limit toward c, or to swing back and forth between $+c$ and $-c$. Both are read here as signs that the equation has no stable state to return to, rather than as physics.

3. **The ordinary physical laws must hold.** Energy conservation and causality are not negotiable, and an equation that purchases good behaviour by breaking either has not solved the problem but relocated it. This criterion does the least work in the pages that follow and is the easiest to forget, which is why it is worth writing down. The sign of a radiation-reaction term, for instance, is exactly the kind of thing one may change to remove a runaway without noticing what it does to the energy budget.

These three are the scorecard. Sections 3, 4 and 5 each close by returning to them. Two features of the list are worth noticing before it is used. It is stated entirely in terms of the behaviour of solutions and says nothing about the form an equation should take, which is deliberate, since the strategy of the sections that follow is precisely to alter the form and inspect what comes out. And it asks that a charged particle behave, in these respects, like an ordinary mechanical one. That expectation is reasonable and widely shared, but it is worth flagging as an assumption rather than a deduction.

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

[Terminology: throughout these notes "Lorentz–Dirac" (LD, also called Abraham–Lorentz–Dirac) refers to the classical equation of motion of a point charge, never to Dirac's relativistic wave equation.]

2.1. *The world-tube momentum–energy balance*

Dirac's advance was a change of question. Lorentz's route, followed in Sec. 1.3, asks what force the charge exerts on itself and answers by summing the retarded action of one charge element on another. That sum has no meaning for a point charge, since the self-field diverges on the world line. Dirac keeps the point charge and declines to ask about the force at all. He asks instead where the energy and momentum go. Whatever the particle radiates must be paid for out of the particle's own energy and momentum. That balance can be struck without ever evaluating a field on the world line. The divergent self-field still enters the calculation, but only through differences, and the differences are finite.

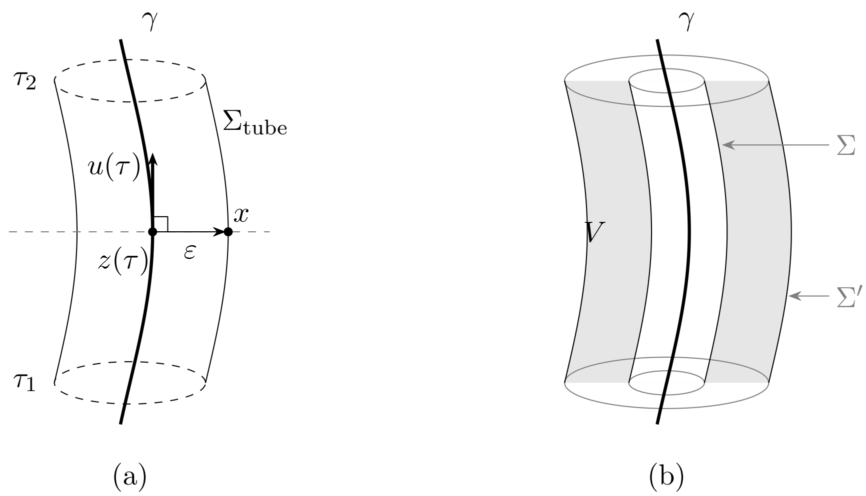


Figure 2. Dirac's construction. *(a)* The tube. At each proper time the point $z(\tau)$ on the world line γ carries a sphere of radius ε, drawn in the hyperplane of simultaneity of $z(\tau)$ — the dashed line — so that the separation $x - z(\tau)$ to a point x of the tube is orthogonal to the four-velocity $u(\tau)$, as in Eq. (15). Dragging that sphere along γ from τ_1 to τ_2 sweeps out Σ_{tube}. The end caps are drawn dashed as a reminder that the integration is carried out over the wall alone; see Sec. 2.5. *(b)* Why the choice of tube does not matter. Two tubes Σ and Σ' enclose the same world line, and the region V between them contains no charge, so the flux of $T^{\mu\nu}$ through the two is the same by Eq. (14). The balance may therefore be read on any surface, including one that keeps well away from the singular world line.

2.1.1. *The tube, and why its shape is immaterial*

Surround the world line γ by a three-dimensional tube Σ. The four-momentum crossing it is

$$\Delta P^\mu = \int_\Sigma T^{\mu\nu}\, d\Sigma_\nu. \tag{13}$$

Let Σ and Σ' both enclose γ, and let V be the region between them. That region contains no charge, so the stress–energy tensor is conserved throughout it, and

$$\Delta P'^\mu - \Delta P^\mu = \oint_{\partial V} T^{\mu\nu}\, d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV = 0. \tag{14}$$

This is the whole reason the strategy beats Lorentz's. It deserves to be stated on its own. A force must be evaluated somewhere, and for a point charge every candidate location is a place where the field is infinite. A balance need not be evaluated anywhere in particular. It may be read off on any surface at all, including surfaces that keep a comfortable distance from the singularity. Equation (14) guarantees that the reading is the same. See Fig. 2(b).

Dirac's tube is the one adapted to the instantaneous rest frame. For a point x on the tube, let $z(\tau)$ be the point of γ satisfying

$$[x^\mu - z^\mu(\tau)]\, u_\mu(\tau) = 0, \qquad [x - z(\tau)] \cdot [x - z(\tau)] = \varepsilon^2, \tag{15}$$

the separation being spacelike. The first condition places x on the hyperplane of simultaneity of $z(\tau)$, so that "at the same moment" means at the same moment for the particle. The second fixes the radius. Together they describe a sphere of radius ε drawn in the instantaneous rest frame and dragged along the world line, sweeping out $\Sigma_{\text{tube}} = \{x(\tau, \varepsilon, \Omega);\ \tau_1 \le \tau \le \tau_2\}$, as in Fig. 2(a).

2.1.2. *The surface element*

Set $c = 1$ for the rest of this subsection and work in the instantaneous rest frame at proper time τ, restoring c at the end. Let n^μ be the unit spacelike vector pointing from $z(\tau)$ to the point x of the tube, so that $x = z(\tau) + \varepsilon n$ with $n \cdot u = 0$ and $n \cdot n = 1$. Three angular integrals are all that will be needed:

$$\oint d\Omega = 4\pi, \qquad \oint n^i \, d\Omega = 0, \qquad \oint n^i n^j \, d\Omega = \frac{4\pi}{3}\delta^{ij}. \tag{16}$$

Differentiate $x = z + \varepsilon n$ along the world line at fixed Ω. The vector n is Fermi–Walker transported, so $\dot{n}^\mu = (a \cdot n)\, u^\mu$. Thus

$$\frac{\partial x^\mu}{\partial \tau} = u^\mu \left(1 + \varepsilon \, a \cdot n\right). \tag{17}$$

Each of the two angular directions contributes a factor ε. Thus

$$d\Sigma_\nu = n_\nu \, \varepsilon^2 \left(1 + \varepsilon \, a \cdot n\right) d\Omega \, d\tau. \tag{18}$$

The factor $1 + \varepsilon \, a \cdot n$ is the one thing in this subsection worth memorizing. It is there because the tube is dragged along a world line that is accelerating. Successive cross-sections are not parallel, and the wall stretches on the side the particle is accelerating towards. It is also easy to write the surface element without it, and everything that distinguishes this calculation from the charged sphere of Sec. 1.3 is contained in it.

2.1.3. *The field near the world line*

Expanded about $z(\tau)$ in the rest frame, the retarded field is

$$\mathbf{E} = \frac{e}{\varepsilon^2}\,\mathbf{n} - \frac{e}{2\varepsilon}\left[\mathbf{a} + (\mathbf{a}\cdot\mathbf{n})\mathbf{n}\right] + \frac{2e}{3}\,\dot{\mathbf{a}} + O(\varepsilon), \tag{19}$$

while $\mathbf{B}$, which vanishes for a charge at rest, begins only at $O(\varepsilon^{-1})$. The surface element carries ε^2, so a product of two fields must reach $O(\varepsilon^{-3})$ to leave anything divergent behind. Since $\mathbf{BB}$ is only $O(\varepsilon^{-2})$, the magnetic field cannot contribute to the divergent term at all, and the whole of the next subsubsection is an exercise with $\mathbf{E}$. The four-momentum leaving the tube is

$$\frac{dP^\mu}{d\tau} = -\oint T^{\mu\nu} n_\nu \, \varepsilon^2 \left(1 + \varepsilon \, a \cdot n\right) d\Omega, \qquad T^{\mu\nu} = \frac{1}{4\pi}\left[F^{\mu\alpha}F^{\nu}{}_{\alpha} - \tfrac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}\right]. \tag{20}$$

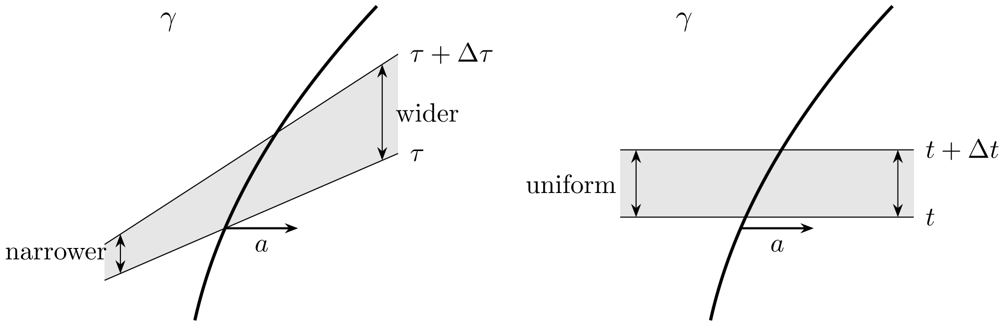
2.1.4. *The divergent term, and what becomes of the* 4/3

Two products of Eq. (19) reach the required order. Take the Coulomb term against itself first.

$$T^{ij} n_j \Big|_{\mathrm{CC}} = \frac{1}{4\pi}\frac{e^2}{\varepsilon^4}\left[n^i n^j - \tfrac{1}{2}\delta^{ij}\right] n^j = \frac{e^2}{8\pi\varepsilon^4}\, n^i. \tag{21}$$

At order ε^{-2} this integrates to nothing, since $\oint n^i d\Omega = 0$. An isotropic Coulomb field pushes equally in every direction. Against the tilt factor it survives:

$$-\oint \frac{e^2}{8\pi\varepsilon^4}\, n^i \, \varepsilon^3 (a \cdot n) \, d\Omega = -\frac{e^2}{8\pi\varepsilon} \cdot \frac{4\pi}{3} a^i = -\frac{e^2}{6\varepsilon}\, a^i. \tag{22}$$



(a) Dirac (b) Lorentz

Figure 3. Where the 4/3 goes. Both panels show the same accelerating world line γ and the same slab of it, and differ only in what is being called "at the same moment". *(a)* Dirac's tube is built from the particle's own succession of rest frames. Because the velocity changes along γ, successive slices are not parallel: they fan open on the side the particle is accelerating towards, and the slab is wider there. That fanning is the factor $1 + \varepsilon\, a \cdot n$ of Eq. (18). *(b)* Collecting field momentum at constant time in one chosen frame gives parallel slices and a slab of uniform thickness, with no such factor. The wedge by which the two shaded regions differ is worth exactly one third of the electromagnetic mass, which is the whole of the discrepancy between Eqs. (4) and (24).

The second product is the Coulomb term against the acceleration field, which is the one a sphere would have too:

$$-\oint \frac{1}{4\pi}\left[-\frac{e^2}{2\varepsilon^3}a^i - \frac{e^2}{2\varepsilon^3}(a\cdot n)n^i\right]\varepsilon^2 d\Omega = \frac{e^2}{2\varepsilon}\left(1+\tfrac{1}{3}\right)a^i = \frac{2e^2}{3\varepsilon}\,a^i. \tag{23}$$

Writing $U_0 = e^2/2\varepsilon$ for the electrostatic energy of a shell of radius ε, the coefficient here is $4U_0/3$: the same 4/3 the sphere produced in Sec. 1.3. It is arrived at in the same way, by ignoring the tilt.

Adding Eqs. (22) and (23),

$$\left.\frac{dP^i}{d\tau}\right|_{\mathrm{div}} = \left(\frac{2}{3}-\frac{1}{6}\right)\frac{e^2}{\varepsilon}\,a^i = \frac{e^2}{2\varepsilon}\,a^i = U_0\,a^i. \tag{24}$$

The coefficient is 1, not 4/3, and the two derivations in these notes therefore disagree about the electromagnetic mass: Eq. (4) absorbed $4U_0/3c^2$, while Eq. (24) absorbs U_0/c^2. Neither contains an error, and it is worth being clear about what the disagreement is.

Section 1.3 recorded the 4/3 as the mark of having counted the electromagnetic stress while omitting whatever holds the sphere together. Dirac's calculation has no sphere to hold together and no binding stress to omit, and it returns U_0/c^2. What produces the difference is the tilt factor of Eq. (18), and the tilt factor is a statement about simultaneity. Lorentz gathers the field momentum on a hyperplane of constant time in one chosen frame. Dirac gathers it on a surface assembled from the particle's own succession of rest frames. The gap between those two ways of deciding what counts as "at the same moment" is precisely the missing third.

The 4/3 was never a statement about electrons. It was a statement about a choice of simultaneity, and it disappears as soon as the choice is made covariantly.

2.1.5. *The finite term, and two checks*

The cross of the Coulomb field with the radiative term $\frac{2e}{3}\dot{\mathbf{a}}$ gives, after the first and third pieces of $T^{ij}n_j$ cancel against each other,

$$-\oint \frac{1}{4\pi}\frac{2e^2}{3\varepsilon^2}\,\dot{a}^i\,\varepsilon^2\,d\Omega = -\frac{2e^2}{3}\,\dot{a}^i. \tag{25}$$

The finite pieces not written out are the acceleration field against itself, $\mathbf{B}$ against itself, and the tilt correction to Eq. (23). They contain no $\dot{a}$, and assemble into the term along u^μ that $u \cdot dP/d\tau = 0$ requires. Restoring c,

$$\frac{dP^\mu}{d\tau} = \frac{e^2}{2\varepsilon c^2}\,a^\mu - \frac{2e^2}{3c^3}\left(\dot{a}^\mu - u^\mu a^\beta a_\beta\right), \tag{26}$$

P^μ being the four-momentum carried out of the tube.

Two things can be checked without redoing anything. Contracting Eq. (26) with u_μ, and using $u \cdot u = -1$ and $u \cdot \dot{a} = -a \cdot a$, gives zero, as it must. And the component along u^μ is $(2e^2/3c^3)\,a \cdot a$, the Larmor power. Thus the radiated energy is present with the coefficient that Sec. 1.3.2 obtained from energy balance, by an argument that never mentioned a tube.

What Eq. (26) is not is an equation of motion. It records how much four-momentum the *field* carries out of the tube. Turning that into a statement about the particle requires saying what the particle's own four-momentum is, and nothing in the calculation supplies it. That freedom is the subject of Sec. 2.2.

2.2. *The undetermined four-vector B^μ*

Total four-momentum is conserved, so the particle's own four-momentum changes at the rate at which the external field supplies it, less the rate at which the field carries it out of the tube,

$$\frac{dP^\mu_{\text{mech}}}{d\tau} = F^\mu_{\text{ext}} - \frac{dP^\mu}{d\tau}, \tag{27}$$

with $dP^\mu/d\tau$ given by Eq. (26). Everything on the right is known. The left is not, because nothing computed so far says what P^μ_{mech} is.

This is the bill for the trick of Sec. 2.1.1. The tube integral was arranged to be insensitive to what lies inside it. That insensitivity is exactly why the divergent self-field never had to be evaluated. But the particle lies inside it. The construction is silent about the interior by design, and it is silent about the particle's own momentum for the same reason. So a form has to be put in by hand:

$$P^\mu_{\text{mech}} = m_0 u^\mu + B^\mu, \tag{28}$$

with B^μ undetermined.

One condition is available. Since $u \cdot u = -1$ throughout, $u \cdot a = 0$, and contracting Eq. (27) with u_μ must therefore give an identity. The Lorentz force is orthogonal to u, and $u \cdot dP/d\tau = 0$ was checked in Sec. 2.1.5, and $m_0\, u \cdot a = 0$. What remains is

$$u_\mu \frac{dB^\mu}{d\tau} = 0. \tag{29}$$

This says no more than that whatever force B^μ contributes must be orthogonal to the four-velocity, which is a requirement on any force term in a relativistic equation of motion. It is the only condition the construction imposes.

It is a very weak condition. One scalar equation restricts four functions, so three functions of τ remain free at every instant. Dirac's choice is the simplest solution,

$$B^\mu = k\, u^\mu, \qquad k = \text{const}, \tag{30}$$

for which $dB^\mu/d\tau = k\, a^\mu$ and Eq. (29) holds automatically. Substituting Eqs. (28) and (30) into Eq. (27) with Eq. (26),

$$\left(m_0 + k + \frac{e^2}{2\varepsilon c^2}\right) a^\mu = F^\mu_{\text{ext}} + \frac{2e^2}{3c^3}\left(\dot{a}^\mu - u^\mu a^\beta a_\beta\right), \tag{31}$$

and the entire content of Dirac's choice is visible in the bracket: k appears only added to m_0 and to the divergent $e^2/2\varepsilon c^2$. Choosing $B^\mu = k u^\mu$ is choosing to add nothing to the dynamics. It shifts a mass, and the mass is not observable separately in any case, since only the sum

$$m = m_0 + k + \frac{e^2}{2\varepsilon c^2} \tag{32}$$

appears. Holding m finite as $\varepsilon \to 0$ requires $m_0 + k \to -\infty$, which should be recorded plainly. The bare mass is negatively infinite, and the observed mass is the residue of a cancellation between two quantities. Neither quantity is separately meaningful.

Two things follow, and they point in opposite directions.

The first is that the derivation has not determined the equation of motion. It has determined it *given* Eq. (30), and Eq. (30) was chosen for simplicity, not derived. Any

$$B^\mu = P_0\, u^\mu + P_1\, a^\mu + P_2\, \dot{a}^\mu + \cdots \tag{33}$$

whose derivative is orthogonal to u is equally admissible, and each choice yields a different equation of motion. It is generally of higher order, since each extra term carries another derivative. Dirac himself put it as a choice of the simplest form rather than as a result. Section 5 takes that door seriously and walks through it.

The second is that the freedom is less mysterious than the notation makes it look, because a particular choice of B^μ is nothing but a choice of what to call the particle's momentum. Suppose one posits, instead of Eq. (30),

$$P^\mu_{\text{mech}} = m_0 u^\mu + C\, \frac{e^2}{c^3} a^\mu \tag{34}$$

and asks what C must be if the field side is to contain only the irreversible radiated momentum. In that bookkeeping the Schott piece has been moved from the field momentum into P^μ_{mech}. Contracting the source-free balance with u_μ, the mechanical side gives $u \cdot (m_0 a + Ce^2\dot{a}/c^3) = -Ce^2\, a \cdot a/c^3$. The remaining radiative field piece contributes $-\frac{2}{3}e^2 a \cdot a/c^3$, so the two cancel only if

$$C = -\tfrac{2}{3}. \tag{35}$$

So orthogonality does fix the coefficient, once the *form* has been assumed. This is the same situation as before, wearing different clothes. What is worth noticing is the object it fixes. The extra piece of the momentum is $-\frac{2}{3}e^2 a^\mu/c^3$, which is the Schott momentum. It is the same quantity that appeared in Sec. 1.3.2 as the boundary term $(2e^2/3c^3)\,\dot{\mathbf{v}}\cdot\mathbf{v}$ that the energy-balance derivation was permitted to discard. It was discarded there by assuming periodic motion. Here it cannot be discarded, and it has to be carried as part of what the particle owns.

2.3. *The equation*

Absorbing the mass as in Eq. (32), Eq. (31) is the Lorentz–Dirac equation,

$$\boxed{ma^\alpha = F^\alpha_{\text{ext}} + \frac{2e^2}{3c^3}\left(\delta^\alpha{}_\beta + u^\alpha u_\beta\right)\dot{a}^\beta} \tag{LD}$$

the projector $\delta^\alpha{}_\beta + u^\alpha u_\beta$ being the compact way of writing $\dot{a}^\alpha - u^\alpha a\cdot a$, and enforcing $u\cdot a = 0$ term by term.

The reaction force has two pieces and they are not the same kind of thing. The piece $\frac{2e^2}{3c^3}\dot{a}^\alpha$ is the Schott term. It is a total derivative, it changes sign with $\dot{a}$, and it is not radiation. It is momentum lent to the near field and returned. The piece $-\frac{2e^2}{3c^3}u^\alpha a\cdot a$ points along the four-velocity, never changes sign, and is exactly the Larmor power carried off, as Sec. 2.1.5 verified. Only the second is irreversible. Almost every puzzle in Sec. 3 is a puzzle about the first. The radiated power and the work done by the reaction force fail to match instant by instant. Uniform acceleration radiates while feeling no reaction at all.

It is worth stating what this derivation did and did not buy, since the same equation was very nearly written down in Sec. 1.3 by Landau and Lifshitz's two-line covariantization, Eq. (8). What is new is not the form. Covariance and $u\cdot a = 0$ fix the form almost by themselves, and the coefficient $2e^2/3c^3$ was already fixed by the nonrelativistic limit. What is new is, first, that no extended body was needed and therefore no binding stress was omitted, which is why the electromagnetic mass came out as U_0/c^2 rather than $4U_0/3c^2$. Second, the freedom left over has been made explicit rather than hidden, in the shape of Eq. (33). A derivation that tells you exactly which of its conclusions were choices is worth more than one that reaches the same equation silently.

What the derivation did not buy is any improvement in the behaviour of the solutions. Equation (LD) still contains $\dot{a}$, so it is still of third order in the position, and Sec. 1.4 already showed what that implies. Section 3 works out the consequences.

2.4. *The other route: separating the singular part*

There is a second way through, and it has the opposite character. Dirac's route declines to ask what force the charge exerts on itself and audits the momentum instead. This one asks about the force after all, but subtracts, before asking, the part of the self-field that makes the question meaningless.

The difficulty being addressed is the one stated in Sec. 1: at the electron both the potential and the field tensor are infinite, so the Lorentz force expression cannot be applied to the self-field. The proposal is to split the infinite self-field into an "irrelevant" singular part, which is discarded, and a finite "effective" part, and then to apply the Lorentz force law to the effective part alone, in the ordinary way. The discarded piece is called the singular term, and the procedure is the method of separating the singular part.

The split that works is between the advanced and retarded potentials. The advanced potential A_{adv} violates causality and is not a candidate for the physical field, but it is a perfectly good solution

Figure 4. What the retarded and advanced potentials are, geometrically. The light cone through a field point x meets the world line twice: once in the past, at $z(u)$, and once in the future, at $z(v)$. The first gives $A_{\rm ret}$, the second $A_{\rm adv}$. *(a)* For x well off the world line the two events are far apart, and only the retarded one is causally admissible. *(b)* As x is brought towards the world line the two roots approach each other and both approach the same event. Both potentials diverge there, at the same rate and with the same coefficient, because both are being sourced by the same piece of the same world line; that is why the singular parts cancel in the difference $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ and survive undiminished in the sum $\frac{1}{2}(A_{\rm ret} + A_{\rm adv})$. Equation (36) splits them where it does for this reason and no other.

of $\Box A^\alpha = -4\pi j^\alpha$, and near the world line its singular behaviour is identical to that of $A_{\rm ret}$. In the difference the singular behaviour cancels. What is kept and what is thrown away are therefore

$$A^\alpha_{(R)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} - A^\alpha_{\rm adv}\right), \qquad A^\alpha_{(S)} = \tfrac{1}{2}\left(A^\alpha_{\rm ret} + A^\alpha_{\rm adv}\right), \tag{36}$$

the first being a difference of two solutions of the inhomogeneous equation and hence a solution of the homogeneous Maxwell equations, regular on the world line, with a finite Lorentz force $f^\alpha = e\,F^{(R)\alpha}{}_\beta u^\beta$.

That this reproduces Dirac's answer is the content of the equivalence theorem:

> The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

Part of the check is already in hand. The finite term computed in Sec. 2.1.5 is precisely e times $\frac{1}{2}(F_{\rm ret} - F_{\rm adv})$ evaluated on the world line, which is why the same $2e^2/3c^3$ appeared there. The agreement of the two routes is not an accident, but neither is it obvious: one of them never evaluates a field at the particle and the other does nothing else.

Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004). The version reproduced here is deferred to Sec. 4.1, and it is worth saying why it gets a section of its own rather than a footnote. Its value is not that it confirms Eq. (LD) a third time. It is that carrying the proof out makes visible something the statement conceals: that Eq. (36) is a *choice* of which combination to discard, and that the choice was made on grounds of convenience. Section 4 relaxes it, replacing Eq. (36) by a two-parameter family, and everything in that section descends from the observation that the split is not forced.

2.5. *The end caps, and where the freedom in B^μ comes from*

One reservation about the derivation remains to be entered, and it turns out not to be independent of the freedom found in Sec. 2.2.

The region V occupied by the tube between τ_1 and τ_2 is bounded by the wall $\Sigma_{\rm tube}$ and by two spacelike end caps $C(\tau_1)$ and $C(\tau_2)$, the discs closing the tube at each end. Conservation applies to the whole boundary, not to part of it:

$$\int_{C(\tau_2)} T^{\mu\nu} d\Sigma_\nu - \int_{C(\tau_1)} T^{\mu\nu} d\Sigma_\nu + \int_{\Sigma_{\rm tube}} T^{\mu\nu} d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu}\, dV. \tag{37}$$

Both the calculation of Sec. 2.1 and Poisson's shorter version (E. Poisson 2004) evaluate the third term only. The 2006 text records this and does not pursue it.

The two cap integrals are not incidental. Each is the four-momentum of the field contained inside the tube at one end,

$$P^\mu_{\rm field}(\tau) = \int_{C(\tau)} T^{\mu\nu} d\Sigma_\nu, \tag{38}$$

so what Eq. (37) says is that the wall integral measures the change in the field momentum *inside* the tube, and not the change in the particle's momentum. Those are different quantities, and separating them requires knowing how the contents of the tube divide between the particle and its field.

Which is exactly what Sec. 2.2 had to put in by hand. If Eq. (38) could be evaluated, $P^\mu_{\rm mech}$ would follow from it rather than being posited, Eq. (28) would not be needed, and the four-vector B^μ would have no freedom left in it. The arbitrariness in B^μ is not a second defect standing alongside the missing caps. It is the missing caps, seen from the side of the equation of motion.

The caps are omitted because they diverge. On a cap the Coulomb energy density goes as e^2/r^4 against a volume element $r^2 dr$, and the momentum density, carrying one factor of $\mathbf{B} \sim 1/r$, as e^2/r^3; the two radial integrals

$$\int_0 \frac{dr}{r^2}, \qquad \int_0 \frac{dr}{r} \tag{39}$$

diverge at the lower limit, the second only logarithmically but no less fatally. The field momentum inside the tube is infinite, which is why it cannot be apportioned, which is why a form for $P^\mu_{\rm mech}$ has to be assumed.

This closes the account of Sec. 2.1 in a way worth stating plainly. The virtue of the tube construction and its cost are the same property. It works because it is insensitive to what lies inside it, and it is for that reason unable to say anything about what lies inside it. The divergence was not removed. It was moved to a place where it could be ignored, and the price of ignoring it is one undetermined four-vector.

A further consequence of the same divergence is left aside here. The stress–energy tensor of the electron's own field curves spacetime, and a quantity that diverges on the world line curves it without bound, so the particle is not in Minkowski space and general relativity has nothing to say about the region where the problem lives. The 2006 text raises this in its Sec. 4 and builds an "effective metric" proposal on it; that material is not covered in these notes.

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

Equation (LD) has been derived twice and its freedoms accounted for. This section asks what it says, and the answer is that it says very nearly everything the Abraham–Lorentz equation said in Sec. 1, in covariant dress. That is not a small disappointment. Dirac's construction repaired the derivation — no extended body, no omitted binding stress, no 4/3 — and left the disease untouched.

3.1. *One-dimensional reduction*

Restrict the motion to one spatial dimension. The normalization $u^\alpha u_\alpha = -1$ reads $\dot{t}^2 - \dot{x}^2 = 1$, which is solved identically by the rapidity $q(\tau)$,

$$\dot{t} = \cosh q, \qquad \dot{x} = \sinh q, \tag{40}$$

so that q carries the whole of the kinematics and the constraint is disposed of once and for all. Differentiating,

$$a^\alpha = \dot{q}\,(\sinh q,\ \cosh q), \qquad \dot{a}^\alpha = \ddot{q}\,(\sinh q,\ \cosh q) + \dot{q}^{\,2}(\cosh q,\ \sinh q). \tag{41}$$

Two things follow at once. First $a\cdot a = \dot{q}^{\,2}$, so the invariant acceleration is just $|\dot{q}|$. Second, and more usefully, the awkward term in Eq. (LD) collapses:

$$\dot{a}^\alpha - u^\alpha\, a\cdot a = \ddot{q}\,(\sinh q,\ \cosh q), \tag{42}$$

because the $\dot{q}^{\,2}$ pieces of $\dot{a}^\alpha$ and of $u^\alpha a\cdot a$ are identical and cancel. Both are proportional to the same vector $(\sinh q, \cosh q)$ that carries a^α, so the two components of Eq. (LD) are not independent: they reduce to a single scalar equation,

$$\dot{q} - \tau_0\, \ddot{q} = \frac{1}{m}\, f(\tau), \tag{43}$$

where $f = eE$ is the scalar force. In the normalization of the 2006 text, $m = e = c = 1$, one has $\tau_0 = 2/3$ and Eq. (43) is written $\dot{q} - \frac{2}{3}\ddot{q} = F^{01}$.

The reduction is worth pausing on, because it makes the structural point of Sec. 1.4 unmissable. Newton's equation in this variable would be $m\dot{q} = f$: rapidity rate proportional to force, no more. What Eq. (43) adds is $\ddot{q}$, and it adds it with a coefficient τ_0 that is fixed, not adjustable. The equation is first order in the acceleration $\dot{q}$, which is to say third order in the position.

3.2. *The free solution and the runaway*

Set $f = 0$. Equation (43) becomes $\ddot{q} = \dot{q}/\tau_0$, whence

$$q = C_1 e^{\tau/\tau_0} + C_2 \qquad \text{and} \qquad \dot{t} = \cosh\!\big(C_1 e^{\tau/\tau_0} + C_2\big)\,, \quad \dot{x} = \sinh\!\big(C_1 e^{\tau/\tau_0} + C_2\big)\,. \tag{44}$$

In the 2006 normalization this is $q = C_1 e^{3\tau/2} + C_2$.

If $C_1 \neq 0$ then $q \to \infty$, and since $dx/dt = \tanh q$, the velocity approaches c. A free electron, in an inertial frame, with nothing whatever acting on it, accelerates itself to the speed of light. The covariant equation has improved on Eq. (AL) only in the sense that the velocity now saturates at c rather than growing without bound; the world line is no less absurd.

If $C_1 = 0$ then q is constant, the electron moves uniformly, and Newton's first law is recovered. This is the physical solution, and the difficulty is the one already met in Sec. 1.4: nothing selects it. The three constants of a third-order equation are fixed by three pieces of data, and position and velocity supply only two. The third is $\dot{q}(0)$, the initial acceleration, which the physical situation does not determine and which the theory offers no way of measuring; and the only value of it that does not lead to disaster must be imposed from outside.

The impulse problem sharpens this exactly as it did before. With $f/m = \kappa\,\delta(\tau)$ and the electron moving uniformly beforehand, integrating Eq. (43) across the pulse and solving on either side gives, for $\tau > 0$,

$$\dot{q} = -\frac{\kappa}{\tau_0}\, e^{\tau/\tau_0}, \tag{45}$$

so the acceleration grows without bound after the blow instead of returning to zero. The problem Sec. 1.4 found in Lorentz's equation is untouched by Dirac's. The 2006 text notes that Zhang's treatment of the single and double pulse (Z. Zhang 1979) produces solutions of the same form.

3.3. *Dirac's asymptotic condition and the price paid*

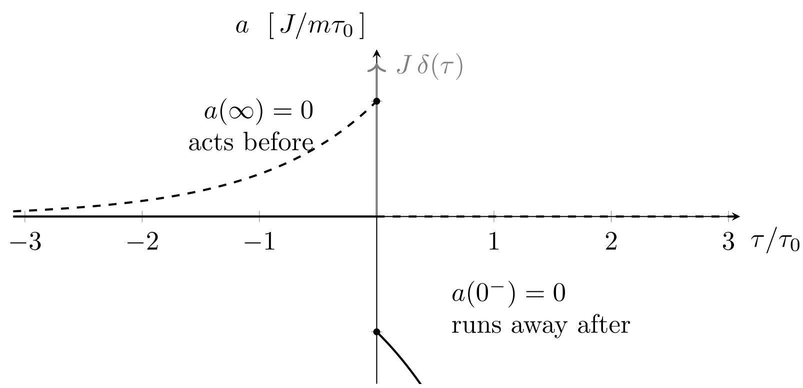


Figure 5. The choice Dirac's asymptotic condition makes. Both curves solve the same equation with the same impulse, and differ only in which end condition is imposed. Fixing the acceleration to vanish before the blow, as an initial-value problem would, gives the solid curve: nothing happens until $\tau = 0$ and the particle then runs away, as in Fig. 1(b). Fixing it to vanish at $\tau \to +\infty$ instead gives the dashed curve: the runaway is gone, the particle ends in uniform motion with exactly the velocity change J/m that the impulse should produce, and it acquires that velocity *before* the impulse arrives, over a time of order τ_0. The two branches are reflections of each other through the origin. There is no third branch, and no choice that avoids both.

Dirac's response was not to modify the equation but to change where the data are supplied. Instead of prescribing the state at some initial instant, impose

$$\dot{q}(\tau) \longrightarrow 0 \qquad \text{as} \qquad \tau \to \pm\infty, \tag{46}$$

that is, require the acceleration to die at both ends of the world line. For the impulse problem this fixes the free constant uniquely, and the solution turns out to be

$$\dot{q} = \frac{\kappa}{\tau_0}\, e^{\tau/\tau_0} \quad (\tau < 0), \qquad\qquad \dot{q} = 0 \quad (\tau > 0). \tag{47}$$

This works, and it works well. The runaway is gone. The velocity change delivered is $\int_{-\infty}^{0} \dot{q}\, d\tau = \kappa$, exactly what an impulse of that strength should produce, so the final state is the correct one. Compared with Eq. (45) the whole pathology has been reflected through $\tau = 0$, as Fig. 5 shows.

And reflecting it through $\tau = 0$ is precisely the trouble. The electron is accelerating at $\tau < 0$. It begins to move before the force reaches it, on a timescale τ_0, and it begins to move in the direction the force will push it, by the amount the force will deliver. It anticipates. Two objections follow, and the 2006 text states both.

The first is procedural. The condition $\dot{q}(0)$ is no longer required — it has been replaced by conditions at $\tau = \pm\infty$. But the motions one actually wants to describe occupy finite stretches of time, and it is a strange theory in which settling what an electron does over a microsecond requires knowing what it will be doing infinitely far in the future. The initial-value problem has not been solved; it has been exchanged for a boundary-value problem whose boundaries are not accessible.

The second is physical. Pre-acceleration is a violation of causality of a particularly naked kind: the effect is not merely correlated with the cause, it precedes it, and it precedes it by an amount that carries information about the cause. In defence one may say that $\tau_0 \sim 10^{-24}$ s is far below any timescale on which classical electrodynamics could be trusted, and that is true; but it is a statement about where the theory should be abandoned, not a resolution of the problem inside it.

What Eq. (46) buys, then, is criterion 2 of Sec. 1.5, and what it spends is criterion 3. That trade is the one fixed feature of this subject, and it will be worth remembering when Secs. 4 and 5 try to arrange a better bargain.

3.4. *Energy balance and the Schott term*

The reaction force in Eq. (LD) has two pieces, and Sec. 2.3 distinguished them by their character. Their effect on the energy budget is what makes the distinction matter. Care is needed here, because a four-force is always orthogonal to u^α. Contracting it with u_α therefore gives zero identically. It is not a power.

To speak about energy, choose an inertial observer with unit time direction n^α. The energy associated with a four-vector P^α is $E = -P\cdot n$. Write the reaction four-force as $R^\alpha = m\tau_0(\dot{a}^\alpha - u^\alpha a\cdot a)$. Its first piece is a total proper-time derivative,

$$m\tau_0\,\dot{a}^\alpha = \frac{d}{d\tau}\left(m\tau_0\,a^\alpha\right), \tag{48}$$

and the quantity being differentiated is the Schott momentum already met in Sec. 2.2. Its second piece, $-m\tau_0\,u^\alpha a\cdot a$, is the radiated four-momentum, $-P_{\rm rad}u^\alpha/c^2$ with $P_{\rm rad} = (2e^2/3c^3)\,a\cdot a$ the Larmor power in the instantaneous rest frame. The energy balance over an interval is therefore

$$\underbrace{\int_{\tau_1}^{\tau_2} -R\cdot n\,d\tau}_{\text{energy supplied by the reaction}} = \underbrace{\Big[-m\tau_0\,a\cdot n\Big]_{\tau_1}^{\tau_2}}_{\text{change in Schott energy}} - \underbrace{\int_{\tau_1}^{\tau_2} P_{\rm rad}(-u\cdot n)\,d\tau}_{\text{radiated energy seen by } n}, \tag{49}$$

which is Eq. (6) again, now written with an observer specified and with the term that Sec. 1.3.2 was allowed to discard carried explicitly.

The radiated power and the work done by the reaction force therefore do not match instant by instant. They match only when the Schott term contributes nothing over the interval, which happens for periodic motion, or when the motion begins and ends with the same value of $a\cdot n$. This is the same condition, and the same evasion, as in Sec. 1.3.2. The difference is that there it was a licence to simplify a derivation, and here it is a statement about what the equation actually asserts. Between one instant and the next, the energy the particle loses and the energy the field carries away are not the same quantity. The discrepancy is stored in, and returned by, the near field.

3.5. *Hyperbolic motion*

The sharpest form of the difficulty is uniformly accelerated motion. Take

$$z^\alpha(\tau) = \frac{1}{\alpha}\left(\sinh\alpha\tau,\ \cosh\alpha\tau\right), \tag{50}$$

a hyperbola in spacetime, with constant proper acceleration α. In the rapidity variable this is simply $q = \alpha\tau$, so $\dot{q} = \alpha$ and $\ddot{q} = 0$. By Eq. (42) the reaction force vanishes identically,

$$\dot{a}^\alpha - u^\alpha\, a \cdot a = 0, \tag{51}$$

which is easily checked directly, since $\dot{a}^\alpha = \alpha^2 u^\alpha$ and $a \cdot a = \alpha^2$.

So Eq. (LD) says that a uniformly accelerated charge feels no radiation reaction at all, and follows exactly the world line an uncharged particle of the same mass would follow under the same applied force. Yet $a \cdot a = \alpha^2 \neq 0$, so the Larmor formula says it radiates, steadily and forever, at the rate $(2e^2/3c^3)\alpha^2$. Something is paying for that radiation and it is not the applied force, which is doing exactly the work required to produce the motion and no more.

Equation (49) says where the accounting goes. In the inertial frame with $n^\alpha = (1, 0)$, the Schott energy is $E_{\text{Schott}} = -m\tau_0 a \cdot n = m\tau_0\alpha \sinh \alpha\tau$. Its change over a finite interval is exactly $\int P_{\text{rad}}(-u \cdot n)d\tau$. The reaction force vanishes, so the left side of Eq. (49) is zero, and the radiated energy is paid for by the change in the Schott energy of the near field.

This is a consistent account, but it is not a comfortable one. The result depends on a near-field energy that grows without bound for eternal hyperbolic motion, and it ties the energy emitted in a finite interval to the history of a solution that has existed forever. The point to carry forward is not that hyperbolic motion is a contradiction. It has been discussed for a century and there are consistent ways of reading it. The point is that the trouble sits, once again, in the Schott term. Every awkward feature of Eq. (LD) traces back to the same $\dot{a}^\alpha$: the third order and the runaway, the pre-acceleration that removing the runaway costs, the failure of the energy budget to close instant by instant, and now a radiating particle with nothing pushing back on it.

3.6. *Scorecard, and the fork in the road*

Against the three criteria of Sec. 1.5:

1. **Solvable as an ordinary initial-value problem?** No. The equation is third order, so it needs three pieces of data, and the physical situation supplies two. The third is the initial acceleration, and every value of it but one leads to a runaway.

2. **Stable under a small disturbance?** Not as an initial-value problem: Eq. (45) runs away. Yes, if Eq. (46) is imposed instead, but then the problem is no longer an ordinary initial-value problem.

3. **Conservation and causality?** Causality fails as soon as Eq. (46) is imposed, by Eq. (47). Energy is conserved, but only in the integrated sense of Eq. (49), and Sec. 3.5 shows how uncomfortable that can get.

The pattern is worth stating in one line, because it is what the rest of these notes is a response to. As an initial-value problem, LD keeps the equation local and causal but is unstable. With Dirac's asymptotic condition it becomes stable, but only by becoming a boundary-value problem with pre-acceleration. No choice of boundary condition satisfies all three criteria, because the difficulty is not in the boundary conditions. It is in the $\dot{a}^\alpha$.

Two ways of attacking that term present themselves, and they are the subjects of the two sections that follow.

- *Sideways.* Keep the structure of the reaction term and open up its coefficient. Section 4 does this, arriving at it from the observation of Sec. 2.4 that the retarded–advanced split of Eq. (36) was a choice.
- *Upward.* Keep the coefficient and raise the order of the equation by enlarging B^μ beyond Dirac's simplest choice, along the lines left open by Eq. (33). Section 5 does this.

This contrast is the spine of the notes. One of the two routes will turn out to survive its own motivation and the other will not, but that is not visible yet, and both are pursued on their own terms.

4. SPLITTING OFF THE SINGULAR PART: THE λ–κ FAMILY

This is the sideways branch of the fork of Sec. 3.6: keep the order of the equation and open up the coefficient of the reaction term. The door was left open in Sec. 2.4. There the self-field was split into a singular part to be discarded and an effective part to be kept,

$$A^\alpha_{(R)} = \tfrac{1}{2}\,(A^\alpha_{\rm ret} - A^\alpha_{\rm adv})\,, \qquad A^\alpha_{(S)} = \tfrac{1}{2}\,(A^\alpha_{\rm ret} + A^\alpha_{\rm adv})\,,$$

the decomposition being Dirac's (P. A. M. Dirac 1938), and the modern form of the same idea being the singular–regular split of S. Detweiler & B. F. Whiting (2003). It was said in Sec. 2.4 that this split is a choice made on grounds of convenience rather than a result. This section makes good on that, by first proving that the choice reproduces Dirac's equation (Sec. 4.1), then asking whether the discarded part is really inert (Sec. 4.2), and then relaxing the split into a two-parameter family and following where the parameters lead.

Of the two branches of the fork this is the one to watch. Whatever becomes of the argument that motivates it — and Sec. 4.2 will have to be read with some care — the equation it produces is Eq. (LD) with the coefficient of the reaction term replaced by a free parameter, and that object has outlived everything else in D. Zhang (2006).

4.1. *The equivalence theorem*

> The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

The theorem is not new. Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004), and neither is short. The one reproduced below is the elementary argument of D. Zhang (2006), given there precisely because the standard routes are laborious. It is worth setting out not only because it is quicker but because of what becomes visible while it is being carried out.

Start from the retarded potential of the charge (see, e.g., J. D. Jackson 1999, ch. 14),

$$A^\alpha_{\rm ret}(t,\mathbf{x}) = \int \frac{j^\alpha(t - |\mathbf{x}-\mathbf{x}'|,\ \mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\, d^3x', \tag{52}$$

and expand the source about the present time rather than the retarded one. Writing $R = |\mathbf{x}-\mathbf{x}'|$ and using $j^\alpha(t-R) = \sum_l \frac{(-R)^l}{l!}\partial_t^l j^\alpha(t)$,

$$A^\alpha_{\rm ret} = \int \sum_{l=0}^{\infty} \frac{(-1)^l}{l!}\, R^{l-1}\, \partial_t^l j^\alpha(t,\mathbf{x}')\, d^3x'. \tag{53}$$

The advanced potential is the same integral with $t+R$ in place of $t-R$, so its expansion is identical except that the factor $(-1)^l$ is absent,

$$A^\alpha_{\rm adv} = \int \sum_{l=0}^{\infty} \frac{1}{l!}\, R^{l-1}\, \partial_t^l j^\alpha(t,\mathbf{x}')\, d^3x'. \tag{54}$$

Everything now follows from the parity of l. In the half-difference the coefficient of the lth term is $\frac{1}{2}[(-1)^l - 1]$, which is zero for every even l and -1 for every odd one: *the even terms cancel and the odd terms survive.* Inserting the point-charge source, $j^0 = e\,\delta(\mathbf{x}-\mathbf{z}(t))$ and $\mathbf{j} = e\,\mathbf{v}\,\delta(\mathbf{x}-\mathbf{z}(t))$, and carrying out the $\mathbf{x}'$ integration against the delta functions,

$$\mathbf{A}_{rr} = -\sum_{l\ \rm odd} \frac{1}{l!}\frac{\partial^l}{\partial t^l}\Big[\mathbf{v}(t)\,|\mathbf{x}-\mathbf{z}(t)|^{\,l-1}\Big], \qquad \Phi_{rr} = -\sum_{l\ \rm odd} \frac{1}{l!}\frac{\partial^l}{\partial t^l}\,|\mathbf{x}-\mathbf{z}(t)|^{\,l-1}, \tag{55}$$

writing A_{rr} for $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$.

The regularity of A_{rr} can now be read off, and it is not luck. (The argument in the next two sentences is not in D. Zhang 2006, which establishes finiteness instead by noting that a difference of two solutions of the inhomogeneous equation solves the homogeneous one. The conclusion is the same; the route below makes it visible one step earlier, and generalizes.) The only term of Eqs. (53) and (54) carrying a negative power of R is $l=0$, which is the Coulomb term; and $l=0$ is even, so it is exactly the term the half-difference removes. Every surviving term has $l \geq 1$, hence a non-negative power R^{l-1}, and nothing in Eq. (55) blows up as $\mathbf{x} \to \mathbf{z}(t)$. This is the algebraic form of what Fig. 4 showed geometrically: the two potentials are sourced by the same piece of the same world line in the same way and differ only in the sign of the retardation, so their singular parts are identical and cancel in the difference.

The rest is computation. With A_{rr} regular one may take $\mathbf{E} = -\nabla\Phi_{rr} - \partial_t \mathbf{A}_{rr}$ and $\mathbf{B} = \nabla \times \mathbf{A}_{rr}$, evaluate them on the world line, and form $\mathbf{F} = e(\mathbf{E} + \mathbf{v}\times\mathbf{B})$. Carrying out the differentiations — the $l=1$ and $l=3$ terms of Eq. (55) are the ones that contribute — gives the three-dimensional form of the Lorentz–Dirac equation, which is the content of Appendices A.1 and A.2 of D. Zhang (2006). The theorem is proved, and the equation of Sec. 2.3 — obtained there from momentum–energy conservation with no field ever evaluated at the particle — has been recovered from a Lorentz force with a particular potential.

What the proof exposes is the thing to carry forward, and this too goes beyond D. Zhang (2006), which arrives at the same condition from the field-theoretic side rather than from the expansion. The cancellation used no property of $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ beyond the relative sign of its two pieces. For a general combination

$$A^\alpha = \lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv}, \tag{56}$$

the $l=0$ term comes with the coefficient $\lambda+\kappa$, so

$$A^\alpha \text{ is regular on the world line} \iff \lambda + \kappa = 0. \tag{57}$$

This is the same statement as "the retained part satisfies the homogeneous Maxwell equations", since $\lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv}$ has source $(\lambda+\kappa) j^\alpha$; the parity argument simply makes it visible one step earlier. And it says that $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ is not singled out. It is one conveniently normalized point on a line of equally regular choices, and Sec. 4.4 is what happens if one moves along that line.

4.2. *Does the singular part act?*

The split of Eq. (36) is legitimate only if the part thrown away does nothing. That is a substantive claim and D. Zhang (2006) examines it rather than assuming it. The examination is the load-bearing step of this section, and it is also the step most in need of checking, so both halves of that are set out here.

Begin with the difficulty that makes the claim non-trivial. Computing the field of $A_{(S)}$ in retarded coordinates and contracting it with the four-velocity gives a four-force whose leading behaviour is

$$f^\alpha = eF^{(S)\alpha}{}_\beta u^\beta = e^2\left[\frac{k^\alpha - u^\alpha}{r^2} + O\left(\frac{1}{r}\right)\right], \tag{58}$$

and k^α does not approach u^α as $r \to 0$: the angle between them is fixed along each generator. So Eq. (58) is an infinite force, going as $1/r^2$, with nothing to balance it. This is the difficulty of Sec. 1 returning in its original form. To make the point picture work one has to delete an unbalanced infinity, and deleting it is not something the theory licenses.

What licenses it, in the modern treatment, is a separate principle: the singular field exerts no force on the particle, all of the self-force coming from the regular field. This is the Detweiler–Whiting principle (S. Detweiler & B. F. Whiting 2003), and D. Zhang (2006) invokes it by name, from Poisson's review (E. Poisson 2004), while describing it frankly as a practical rule standing in for a principle that is missing.

The question is then a sharp one. *Is the Detweiler–Whiting singular field the same object as* $\frac{1}{2}(A_{\rm ret} + A_{\rm adv})$*?* If it is, the split of Eq. (36) is forced and there is nothing to relax. The 2006 text answers no, and the answer is a computation. Building the stress–energy tensor of the symmetric combination,

$$T^{\alpha\beta}_{(S)} = \frac{1}{4\pi}\left[F^{(S)\alpha}{}_\delta F^{\delta\beta}_{(S)} - \tfrac{1}{4}\eta^{\alpha\beta}F^{(S)}_{\mu\rho}F^{\mu\rho}_{(S)}\right], \tag{59}$$

and integrating its flux in the manner of Sec. 2.1 gives, for a unit charge, Eq. (1.2.17) of D. Zhang (2006):

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \tag{60}$$

The first term is the familiar divergent piece proportional to the acceleration, and is absorbed into the mass as in Sec. 2.2. The second is finite and does not vanish. Read as the effect of $A_{(S)}$ on the particle, it says that the symmetric combination is *not* inert, hence is not the Detweiler–Whiting singular field, hence that Eq. (36) is one choice among others. That is the licence Sec. 4.3 uses.

Now the other half. Equation (60) should not be relied on without an independent recomputation, for two reasons.

The first is that it stands against the literature. In flat spacetime the Detweiler–Whiting singular field *is* the half-sum, and the theorem (S. Detweiler & B. F. Whiting 2003) is precisely that it exerts no force. Both statements cannot be true. The 2006 text is aware of the principle and is proposing that the identification fails; the modern position is that it does not.

The second is that the quantity computed may not be the quantity wanted. The force on the particle follows from the balance of the *total* field, and

$$T[F_{\rm ret}] = T\big[F_{(S)}\big] + T\big[F_{(R)}\big] + 2\,T\big[F_{(S)}, F_{(R)}\big]\,, \tag{61}$$

so the flux of the singular field's own stress tensor omits the cross terms. "The momentum carried by the singular field" and "the force the singular field exerts" are different quantities, and Eq. (60) computes the first. Whether the $\frac{2}{3}\dot{a}^\alpha$ survives once Eq. (61) is treated in full, or whether it is an artefact of the algebra in Appendix A.3 of D. Zhang (2006), is not settled here.

What can be said without settling it is stated in Sec. 4.7, and it is worth flagging now so that the reader is not left waiting: the family constructed in the next two subsections does not depend on Eq. (60) for its usefulness, only for its original motivation.

4.3. *The two-parameter family*

Relax the form of the discarded part, but keep the requirement that what remains be built from the same two potentials. Writing the retained field as in Eq. (56),

$$A^\alpha = \lambda A^\alpha_{\rm ret} + \kappa A^\alpha_{\rm adv},$$

the parity decomposition of Sec. 4.1 does the work immediately. The lth term of $A_{\rm ret}$ carries $(-1)^l$ and that of $A_{\rm adv}$ carries 1, so the combination is

$$A^\alpha = (\kappa - \lambda)\sum_{l\ \rm odd} \frac{1}{l!}\,\partial_t^l[\,\cdot\,] \;+\; (\lambda+\kappa)\sum_{l\ \rm even}\frac{1}{l!}\,\partial_t^l[\,\cdot\,]\,, \tag{62}$$

the brackets standing for the same quantities as in Eq. (55). The odd series is the regular one; the even series is the one containing $l = 0$ and hence the whole of the singularity.

Two conditions on (λ, κ) then suggest themselves, and D. Zhang (2006) takes both.

1. **Branch (a), $\lambda + \kappa = 0$.** The even series drops out entirely, so the retained field is regular on the world line. This is Eq. (57), and equivalently the statement that the retained field satisfies the homogeneous Maxwell equations. Section 4.4.
2. **Branch (b), $\lambda - \kappa = 1$.** The *discarded* part is $(1-\lambda)A^\alpha_{\rm ret} - \kappa A^\alpha_{\rm adv}$, and this condition makes retarded and advanced enter it with equal weight. The retained field then keeps a singular part. Section 4.5.

These are two lines in the (λ, κ) plane and they meet at one point,

$$\lambda = \tfrac{1}{2}, \qquad \kappa = -\tfrac{1}{2}, \tag{63}$$

which is Dirac's choice, Eq. (36). So the standard split is the unique member of the family satisfying both conditions at once. That is worth stating, because it explains why $\frac{1}{2}(A_{\rm ret} - A_{\rm adv})$ looks canonical: it is distinguished, just not uniquely by either requirement on its own.

Branch (a) can be settled here, before any equation of motion is solved. Setting $\lambda = -\kappa$ in Eq. (62) leaves $\kappa - \lambda = 2\kappa$, so

$$A^\alpha\big|_{\lambda+\kappa=0} = -2\kappa\, A^\alpha_{rr}, \tag{64}$$

with A_{rr} the Dirac field of Sec. 4.1. The retained field on this branch is nothing but Dirac's, rescaled by -2κ. Field, force and reaction term are all linear in the potential, so the reaction term of Eq. (LD) is rescaled by the same factor and nothing else changes:

$$\frac{2e^2}{3} \;\longrightarrow\; -\frac{4\kappa e^2}{3}, \tag{65}$$

which is Eq. (1.5.1) of D. Zhang (2006), recovered without solving anything. Setting $\kappa = -\frac{1}{2}$ returns the coefficient $2e^2/3$ and with it Eq. (LD), as Eq. (63) requires.

The whole of branch (a) is therefore the Lorentz–Dirac equation with one number changed. Section 4.4 has only to work out what changing it does to the solutions. Branch (b) admits no such shortcut, since the retained field keeps its singular part and the rescaling argument does not apply; it has to be handled numerically, in Sec. 4.5.

4.4. *Branch (a):* $\lambda + \kappa = 0$

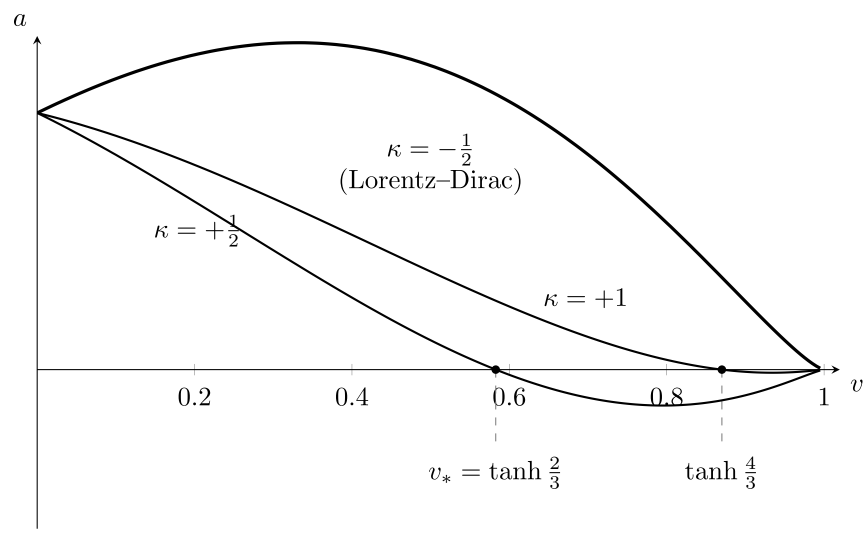


Figure 6. The phase portrait of branch (a), Eq. (68), for $C = 1$. The sign of κ decides everything. For $\kappa < 0$, which includes the Lorentz–Dirac value $\kappa = -\frac{1}{2}$, the bracket grows with v and the curve never returns to the axis: the acceleration is positive at every speed, so nothing stops the particle short of c, and this is the runaway of Sec. 3.2 seen in the (v, a) plane. For $\kappa > 0$ the bracket decreases, the curve must cross the axis, and the crossing at $v_* = \tanh(4\kappa C/3)$ is an attractor: below it the acceleration is positive and above it negative, so a particle released anywhere returns to v_*. The terminal speed is then fixed by C, which is fixed by the applied impulse — force determines velocity rather than acceleration, which is why D. Zhang (2006) calls this Aristotle motion. Only $v > 0$ is shown; the portrait is symmetric under $(v, a, C) \to (-v, -a, -C)$.

Section 4.3 already established what this branch is: the Lorentz–Dirac equation with its reaction coefficient rescaled by -2κ,

$$ma^\alpha = F^\alpha_{\text{ext}} - \frac{4\kappa}{3} e^2 \left(\dot{a}^\alpha - a^\beta a_\beta \, u^\alpha\right), \tag{66}$$

with $\kappa = -\frac{1}{2}$ returning Eq. (LD). What remains is to find out what the rescaling does, and for the free particle in one dimension that can be answered in closed form.

Write $a = \dot{v} = p(v)$, so that $\ddot{v} = p\, dp/dv$ and the equation of motion becomes first order in the (v, a) plane. With no external force Eq. (66) reduces to

$$\frac{dp}{dv} + \frac{3v}{1 - v^2}\, p = -\frac{3}{4\kappa}\sqrt{1 - v^2}, \tag{67}$$

a linear equation whose integrating factor is $(1 - v^2)^{-3/2}$, and which therefore integrates at once to

$$a = \left[-\frac{3}{8\kappa} \ln \frac{1 + v}{1 - v} + C\right] (1 - v^2)^{3/2}. \tag{68}$$

This is Eq. (1.2.26) of D. Zhang (2006). The constant C is fixed by the state the particle is put in — for a particle kicked from rest, by the size of the kick — and κ is the parameter of the family. Everything about this branch can be read off the one curve, and it is drawn in Fig. 6.

The sign of κ decides the whole question, and it decides it by elementary means. As v runs from 0 to 1 the logarithm runs from 0 to $+\infty$. So for $\kappa < 0$ the bracket *increases* without bound, the curve stays above the axis at every speed, and the acceleration is positive everywhere: nothing brings the particle to rest relative to its own motion and it proceeds to c. That is the runaway of Sec. 3.2, and since $\kappa = -\frac{1}{2}$ lies in this range it is also the statement that Eq. (LD) belongs to the bad half of the family.

For $\kappa > 0$ the bracket *decreases*, from C at $v = 0$ towards $-\infty$. It must therefore vanish somewhere, and where it vanishes the acceleration does too. Setting the bracket to zero and using $\ln \frac{1+v}{1-v} = 2 \operatorname{artanh} v$,

$$v_* = \tanh\left(\frac{4\kappa C}{3}\right). \tag{69}$$

Below v_* the acceleration is positive and above it negative, so v_* is not merely a zero but an attractor: a particle released at any speed is carried to it. D. Zhang (2006) identifies the crossing and reads it as the point of re-equilibration; Eq. (69) writes it out.

That is a genuine damping. A particle disturbed from uniform motion settles back to uniform motion, which is what criterion 2 of Sec. 1.5 asks for and what nothing in Secs. 3 or 5 delivers.

It also has a peculiar character, and the 2006 text's name for it is the right one. The speed the particle ends at is set by C, and C is set by the force that was applied. *The force determines the velocity, not the acceleration.* That is Aristotle's picture of motion rather than Newton's, and Eq. (69) is its quantitative form; hence *Aristotle motion*. It is worth being clear that this is not a slip. It is what a damping term of this sign does: the reaction opposes the motion at every speed until the opposition balances the applied force, exactly as a body in a viscous medium reaches a terminal velocity, and Aristotle's mistake was to suppose that the world is always in that regime.

Two things should be entered against this branch before it is left.

The first is that κ is not determined. It is a free parameter, and the theory says only that it must be positive for the behaviour above. The 2006 text is explicit that it would have to be fixed by experiment, which is what its Sec. 5 is about and is not covered in these notes.

The second is more serious and is not raised in the 2006 text at all. Turning κ positive reverses the sign of the reaction term in Eq. (66). The term whose sign was reversed is the one identified in Sec. 2.3 as the radiated four-momentum, $-P_{\mathrm{rad}} u^\alpha / c^2$ with $P_{\mathrm{rad}} \geq 0$ by the Larmor formula. Reversing it does not merely damp the motion; it turns radiative loss into radiative gain, and criterion 3 of Sec. 1.5 was written down with exactly this kind of move in mind. No energy budget for Eq. (66) appears anywhere in the 2006 text, and it should be computed before the $\kappa > 0$ branch is taken to have solved anything. The runaway is certainly gone. Where the energy comes from is an open question.

4.5. *Branch (b):* $1 - \lambda = -\kappa$

[Retarded and advanced parts enter the discarded singular term on equal footing. Solve numerically in κ. Result: $\kappa = -1$ is the watershed. For $\kappa > -1$ the runaway survives; for $\kappa < -1$ a critical velocity v_c appears — a particle starting below v_c stays below it, one starting above decelerates stably. Reproduce the table:]

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

4.6. *An unresolved point*

[Extending branch (b) beyond one dimension runs into $\mathbf{n} \cdot \mathbf{v}$ being undefined at the position of the charge itself. The 2006 text suggested the replacement $\mathbf{n} \cdot \mathbf{v} \to \frac{1}{3}v$ but did not carry it out.]

4.7. *What this branch amounts to*

[Eq. (66) is nothing but the LD equation with the radiation-reaction coefficient $2e^2/3$ replaced by a free parameter $-4\kappa e^2/3$. State this plainly — it is the one result in these notes that outlives its original motivation, whatever becomes of Eq. (60).]

5. THE ELIEZER-TYPE SERIES

This is the upward branch of the fork of Sec. 3.6: keep the coefficient of the reaction term and raise the order of the equation. The door was left open in Sec. 2.2. The tube calculation determines B^μ only through Eq. (29), one scalar condition on four functions; Dirac closed it with the simplest choice, and this section asks what the other choices give.

5.1. *A note on the name*

The equations of this section are not the equation now standardly called the Eliezer equation, and the difference matters enough to state before anything else.

Eliezer's 1947 review (C. J. Eliezer 1947) contains an ansatz for the undetermined vector,

$$B_\mu = -\tfrac{3}{2}P_2\,\dot{v}^2 v_\mu + P_2\,\ddot{v}_\mu, \qquad P_2 = \text{const}, \tag{70}$$

and it is this that the 2006 text generalizes and names after him. What the literature calls the Eliezer, or Eliezer–Ford–O'Connell, equation (C. J. Eliezer 1948; G. W. Ford & R. F. O'Connell 1991) is a different object entirely: it is obtained by *reducing* the order of Eq. (LD) rather than raising it, it is free of runaway solutions, and it is one of the live candidates in present-day radiation-reaction experiments.

The two therefore sit on opposite sides of the fork. Reading the conclusion of this section as a verdict on the Eliezer–Ford–O'Connell equation would invert it exactly.

5.2. *Expanding B^μ*

Write the undetermined vector as a series in the derivatives of the four-velocity,

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots\,, \tag{71}$$

the coefficients P_i being undetermined and not necessarily constant. The constraint that shapes the series is the normalization itself. Differentiating $v^\mu v_\mu = -1$ repeatedly gives a chain of identities,

$$\begin{aligned} v_\mu \dot{v}^\mu &= 0, \\ v_\mu \ddot{v}^\mu + \dot{v}_\mu \dot{v}^\mu &= 0, \\ v_\mu \dddot{v}^\mu + 3\dot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(4)\mu} + 4\dot{v}_\mu \dddot{v}^\mu + 3\ddot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(5)\mu} + 5\dot{v}_\mu v^{(4)\mu} + 10\ddot{v}_\mu \dddot{v}^\mu &= 0, \end{aligned} \tag{72}$$

and so on, the coefficients being those of Pascal's triangle because the identities come from the Leibniz rule; the general term is given in Appendix A.4 of the 2006 text.

Two things follow. The odd-numbered derivatives of v_μ can always be traded for lower ones by Eq. (72), so *only the even terms of Eq. (71) survive as independent*, and the coefficient of v_μ collects everything else:

$$B_\mu = \Big\{k - \tfrac{3}{2}P_2(\dot v)^2 - 5P_4\,\dot v_\mu\,\dddot v^{\,\mu} - \tfrac{5}{2}P_4(\ddot v)^2 - \cdots\Big\}v_\mu \;+\; P_2\ddot v_\mu + P_4 v_\mu^{(4)} + \cdots + P_{2i}v_\mu^{(2i)}, \tag{73}$$

which is Eq. (1.3.2) of the 2006 text. Substituting into the balance of Sec. 2.2 gives a generalized equation of motion, its Eq. (1.3.3), carrying one undetermined constant for every even derivative retained.

The parameters cannot be computed from the theory. They would have to be measured, which is a point worth holding onto: each term added to Eq. (71) buys a new free constant and a new order of derivative at the same time. Setting $P_2 = P_4 = \cdots = 0$ returns Eq. (LD), so LD is the first member of the family.

5.3. *Keeping P_2: a third-order equation*

Retain only P_2, so that $B_\mu = P_0 v_\mu + P_2\ddot v_\mu$. The equation of motion acquires one derivative beyond LD, and the one-dimensional reduction of Sec. 3.1 turns it into a single nonlinear equation for the rapidity,

$$\dddot q - B\ddot q + A\dot q - \tfrac{1}{2}\dot q^{\,3} = 0, \qquad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}, \tag{74}$$

k being the retained coefficient. Two features of A and B do real work later. Since $m > 0$ and $e^2 > 0$ and both are divided by the same k, *A and B necessarily have the same sign*, so only the quadrants $A, B > 0$ and $A, B < 0$ are available. And their ratio is not free either:

$$\frac{B}{A} = \frac{2e^2}{3m} = \tau_0 \tag{75}$$

in units with $c = 1$, which is Eq. (9) again. For an actual electron τ_0 is minute, so the physically interesting corner is $B \ll A$.

Equation (74) has no general analytic solution, so it has to be examined numerically, and that raises the question of what initial data to use. The 2006 text answers it in the way that makes physical sense: take the trivial solution $\dot q = 0$, a uniformly moving electron, disturb it with an impulse, and use the state just after the impulse. So one needs to know what an impulse does to a third-order equation. With $\dddot q - B\ddot q + A\dot q - \frac{1}{2}\dot q^{\,3} = s\,\delta(\tau)$, the answer is that $\ddot q$ *jumps* and q, $\dot q$ are unaffected: if $\dot q$ jumped then $\ddot q$ would contain a δ and $\dddot q$ a δ', which the right-hand side does not have. Estimating from $\dddot q - B\ddot q = s\,\delta(\tau)$,

$$\ddot q = \begin{cases} 0, & \tau < 0, \\ s\,e^{B\tau}, & \tau > 0, \end{cases} \tag{76}$$

so the jump is s, independent of A and B: the impulse sets the second derivative and nothing else. The pattern of Sec. 3.2 has moved up one rung. Newton's impulse moves the velocity, LD's moves the acceleration, this one moves the derivative of the acceleration.

Before the scan, the 2006 text records what would count as success, and the second item is sharper than anything in Sec. 1.5:

1. After the impulse the electron should return to a stable state — not necessarily uniform motion, but at any rate not always reaching c.
2. For particular (A, B) one might accept the electron reaching c, *provided there is a critical impulse*: below some threshold the electron must not reach c. Equation (LD) has no such feature, since under it any impulse whatever leads to c.
3. The scan is a theoretical exercise. If the equation held in nature there would be one value of (A, B), not a family.

The result of the scan is uniformly negative. For every sampled (A, B), in both admissible quadrants and for a range of initial $\ddot{q}(0)$, the electron accelerates itself to the speed of light. In the quadrant $A, B < 0$ it does so monotonically, behaving very much as it does under LD. In the quadrant $A, B > 0$ the behaviour splits: for $A \sim B$ and for $A \ll B$ it again resembles LD, but for $A \gg B$ — which by Eq. (75) is the physically relevant corner — something worse appears. The velocity swings between $+c$ and $-c$, at large amplitude and high frequency, and goes on doing so. Figures 1-3-1 to 1-3-4 of the 2006 text show $v(\tau)$ for $(A, B) = (2, 1)$, $(3, 1)$, $(10, 1)$ and $(100, 1)$ at $\ddot{q}(0) = 2$.

By criterion 2 of Sec. 1.5 an oscillation between $\pm c$ is not a stable state to return to. So the third-order member of the family fails, and fails worse than the equation it was meant to improve on. Eliezer regarded his equation as an improvement on LD; on the evidence of this scan it is not one.

5.4. *Keeping P_2 and P_4: a fifth-order equation*

Retaining P_4 as well produces Eq. (1.3.9) of the 2006 text, and the one-dimensional reduction gives a fifth-order nonlinear equation,

$$q^{(5)} + \left[C + 5\dot{q}^2\right] \dddot{q} - B\ddot{q} + \left[A + \tfrac{25}{2}\dot{q}^2\right] \dot{q} - \tfrac{3}{2}\dot{q}^5 - \tfrac{1}{2}C\dot{q}^3 = 0, \tag{77}$$

with A, B and C fixed by m, e^2, P_2 and P_4, and $C = P_2/P_4$. As before A and B share a sign, but C does not: its sign is free, and the scan uses that freedom.

The impulse now enters one rung higher again. Since Eq. (77) has no $q^{(4)}$ term, what can jump is $\dddot{q}$ or $q^{(4)}$; working from $q^{(5)} + C\dddot{q} = s\,\delta(\tau)$ gives $\dddot{q}(0) = 0$ and $q^{(4)}(0) = s$, so it is the fourth derivative that the impulse sets. The 2006 text notes that this generalizes: each pair of terms added to Eq. (71) pushes the quantity an impulse acts on two derivatives further from the velocity.

The scan covers twenty-two triples (A, B, C), with initial data $q = \dot{q} = \ddot{q} = \dddot{q} = 0$ and $q^{(4)}(0) = 2$, chosen so that $B \ll A$ in most of them for the reason given in Eq. (75). Three classes of behaviour appear, and this time they are not all bad.

The first two classes need no comment: they are the $\pm c$ oscillation of Sec. 5.3 and the LD runaway respectively. The third is new, and it is worth being precise about what it is, because it is the only place in these notes where raising the order produces something that was not already available.

In this class the electron, after the impulse, never reaches c. That alone distinguishes it from every solution of Eq. (LD) and of Eq. (74). But neither does it return to uniform motion, which is what the λ–κ branch of Sec. 4 delivers. Instead the speed pulsates periodically within a bounded range, indefinitely. The 2006 text's description is the memorable one: the electron behaves very like an oscillator, except that the oscillator itself is not fixed but keeps moving forward, with only the speed varying regularly within a range.

Behaviour	Sampled (A, B, C)
Oscillation between $\pm c$	$(100, 1, 1)$, $(100, 1, -1)$, $(100, 1, -100)$, $(1000, 1, -1000)$, $(-1, -100, 100)$
Monotonic approach to c, as under LD	$(-100, -1, 1)$, $(-100, -1, 100)$, $(-100, -1, -10)$, $(-1000, -1, -1000)$
Neither: bounded periodic pulsation	$(100, 1, 100)$, $(100, 1, 1000)$, $(1000, 1, 100)$, $(1000, 1, 1000)$, $(10000, 1, 1000)$, $(-1, -100, 1)$, $(-1, -1000, 10)$

Table 2. The parameter scan of Eq. (77) reported in the 2006 text, grouped by outcome rather than listed in its original order. All runs start from rest with $q^{(4)}(0) = 2$. The third class is the one that does not occur anywhere in Secs. 3 or 5.3.

This is exactly the case that criterion 2 of Sec. 1.5 was written to admit. The criterion was stated there as requiring a return to uniform motion *or* a regular variation confined to a bounded range of speeds, and the second clause was included for this subsection. So on the scorecard these solutions pass.

Two qualifications belong with that verdict, and the 2006 text supplies both. The label attached to this class in the original table is "undetermined" rather than "reasonable", because every run in the scan used the same initial value $q^{(4)}(0) = 2$; smaller initial data might move some of the failing triples into this class, and the dependence was not explored. And the scan is a sampling of a three-parameter space, not a characterization of it.

A third qualification is mine rather than the 2006 text's. Its table lists $(100, 1, 1)$ twice with two different verdicts, and repeats three other triples; Table 2 keeps one of each. The scan should be redone before any weight is put on the boundaries between the three classes.

5.5. *Scorecard*

Against Sec. 1.5, the family of Eq. (71) does this. Criterion 1 gets worse at every step, monotonically: each retained P_{2i} adds two orders, so the number of initial conditions the physical situation cannot supply grows, and Eq. (76) shows the one datum an impulse does fix retreating further from the velocity as it does. Criterion 2 fails outright at third order and is met by part of the fifth-order parameter space. Criterion 3 is not tested anywhere in this section, which is itself worth recording: the energy budget of these equations was never examined.

So the honest summary is that raising the order bought one thing — the bounded pulsating solutions of Sec. 5.4, which are not available at lower order — at the price of a free constant per step, two orders per step, and no account of energy. Whether that is a bargain the 2006 text could not say, and it closed on the question rather than an answer:

> Would nature really go and choose a more complicated B_μ? And if it did, we truly cannot imagine the source of nature making such a choice.

The question is a good one and it has since been answered, though not in a way that was available in 2006. That answer belongs with the other twenty-years-on material and is not taken up here.

REFERENCES

Abraham, M. 1905, Theorie der Elektrizität, Band II: Elektromagnetische Theorie der Strahlung (Leipzig: Teubner)

Cole, J. M., et al. 2018, Physical Review X, 8, 011020

Detweiler, S., & Whiting, B. F. 2003, Physical Review D, 67, 024025

DeWitt, B. S., & Brehme, R. W. 1960, Annals of Physics, 9, 220

Dirac, P. A. M. 1938, Proceedings of the Royal Society of London A, 167, 148

Eliezer, C. J. 1947, Reviews of Modern Physics, 19, 147

Eliezer, C. J. 1948, Proceedings of the Royal Society of London A, 194, 543

Fermi, E. 1922, Physikalische Zeitschrift, 23, 340

Ford, G. W., & O'Connell, R. F. 1991, Physics Letters A, 157, 217

Gralla, S. E., Harte, A. I., & Wald, R. M. 2009, Physical Review D, 80, 024031

Hobbs, J. M. 1968, Annals of Physics, 47, 141

Jackson, J. D. 1999, Classical Electrodynamics, 3rd edn. (Wiley)

Landau, L. D., & Lifshitz, E. M. 1975, The Classical Theory of Fields, 4th edn. (Butterworth-Heinemann)

Larmor, J. 1897, Philosophical Magazine, 44, 503

Lorentz, H. A. 1909, The Theory of Electrons (Leipzig: Teubner)

Mino, Y., Sasaki, M., & Tanaka, T. 1997, Physical Review D, 55, 3457

Poder, K., et al. 2018, Physical Review X, 8, 031004

Poincaré, H. 1906, Rendiconti del Circolo Matematico di Palermo, 21, 129

Poisson, E. 1999, arXiv e-prints

Poisson, E. 2004, Living Reviews in Relativity, 7, 6

Quinn, T. C., & Wald, R. M. 1997, Physical Review D, 56, 3381

Rohrlich, F. 2007, Classical Charged Particles, 3rd edn. (World Scientific)

Schott, G. A. 1912, Electromagnetic Radiation and the Mechanical Reactions Arising from It (Cambridge University Press)

Spohn, H. 2000, Europhysics Letters, 50, 287

Weinberg, S. 1972, Gravitation and Cosmology: Principles and Applications of the General Theory of Relativity (Wiley)

Wilson, W. 1936, Proceedings of the Physical Society, 48, 736

Zhang, D. 2006, Bachelor's thesis, Department of Astronomy, Nanjing University, Nanjing

Zhang, Z. 1979, Electrodynamics and Special Relativity, 2nd edn. (Beijing: Peking University Press)